2. Bayesian Networks: Graphical Models

CSC3043S Reasoning in AI: Bayesian Networks

Dr Jan Buys

University of Cape Town

[Based in part on slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All materials available at http://ai.berkeley.edu.]



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Probabilistic Inference

- **Probabilistic Inference**: Computing a desired probability from other known probabilities
- Most commonly compute conditional probabilities (from joint probabilities or other conditional probabilities)
- A conditional probability represents the *belief* of the agent given the evidence (variables that are conditioned on)
  - P(on time | no reported accidents) = 0.9
- Probabilities change with new evidence: Observing new evidence cause beliefs to be *updated*
  - P(on time | no accidents, 5 a.m.) = 0.95
  - P(on time | no accidents, 5 a.m., raining) = 0.80

Probabilistic Inference

General setup for probabilistic inference:

- A probabilistic model with the following variables:
  - Evidence variables: $E_1, E_2 \ldots, E_k = e_1, e_2, \ldots, e_k$
  - Query variable: Q
  - Hidden variables: $H_1, H_2, \ldots, H_r$
- The goal is to compute: $P(Q|e_1, e_2, \ldots, e_k)$

Probabilistic Inference

Inference by enumeration:

1. **Select** the probability table entries consistent with the evidence:

$$P(Q, H_1, H_2, \dots, H_r, e_1, e_2, \dots, e_k)$$

2. **Marginalize** the hidden variables:

$$P(Q, e_1, e_2, \dots, e_k) = \sum_{h_1, \dots, h_r} P(Q, h_1, \dots, h_r, e_1, e_2, \dots, e_k)$$

3. **Normalize** the distribution:

$$Z = \sum_{q} P(q, e_1, e_2, \dots, e_k)$$

$$P(Q|e_1, e_2, \dots, e_k) = \frac{1}{Z} P(Q, e_1, e_2, \dots, e_k)$$

Probabilistic Inference

Example: Inference by enumeration

- $P(W)$
  - $P(W) = \sum_{s,t} P(s, t, W)$

- $P(W \mid \text{winter})$
  - $P(W, \text{winter}) = \sum_{t} P(\text{winter}, t, W)$
  - $Z = \sum_{w} P(\text{winter}, w)$

- $P(W \mid \text{winter}, \text{hot})$
  - $Z = \sum_{w} P(\text{winter}, \text{hot}, w)$
  - $P(\text{W}|\text{winter}, \text{hot}) = \frac{1}{Z} P(\text{winter}, \text{hot}, W)$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Probabilistic Inference

Inference by enumeration: Computational Complexity

- For a joint distribution over *n* variables with domain sizes *d*
  - Worst case time complexity is $O(d^n)$
  - Space complexity to store joint distribution is $O(d^n)$

- This is not tractable for a model with a large number of variables
- However, not all variables will interact with one another
  - Make (conditional) independence assumptions

Bayesian Networks

- **Bayesian networks** (BNs) are a framework for describing *complex, joint distributions* using *simple, local distributions* (i.e., conditional probability distributions)
- Also called *directed* **probabilistic graphical models**
- Bayesian networks describe how variables interact locally
  - Local interactions chain together to result in global, indirect interactions

Bayesian Networks

- Example: BN for Car Insurance cost

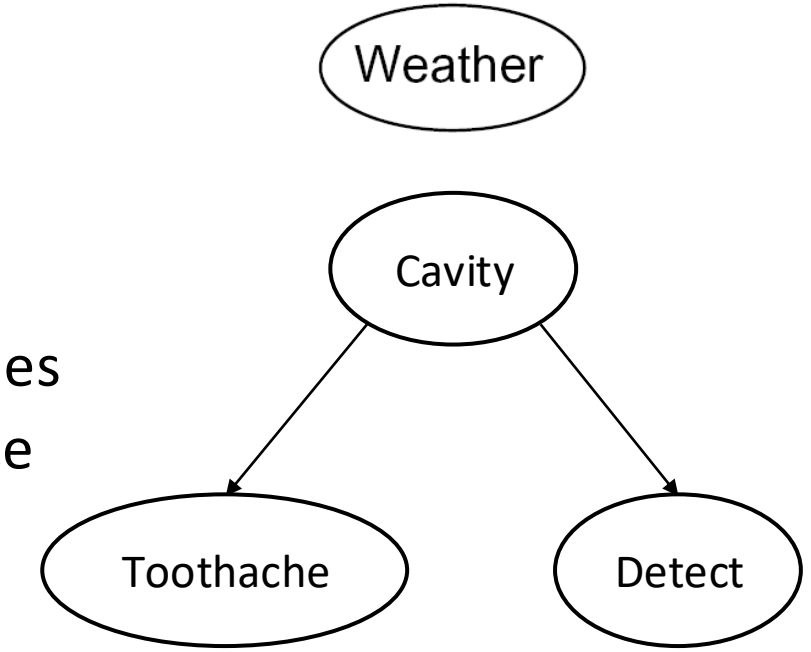


Graphical models

Notation:

- Nodes: Random variables
  - Can be assigned (observed) or unassigned (unobserved)
  - Observed variable's nodes are shared
- Arcs: Interactions between variables
  - Intuitively: Indicate direct influence between variables
  - Formally: Encodes (lack of) conditional independence
  - The arcs are directed: Imagine that they indicate direct causation (in general they don't)

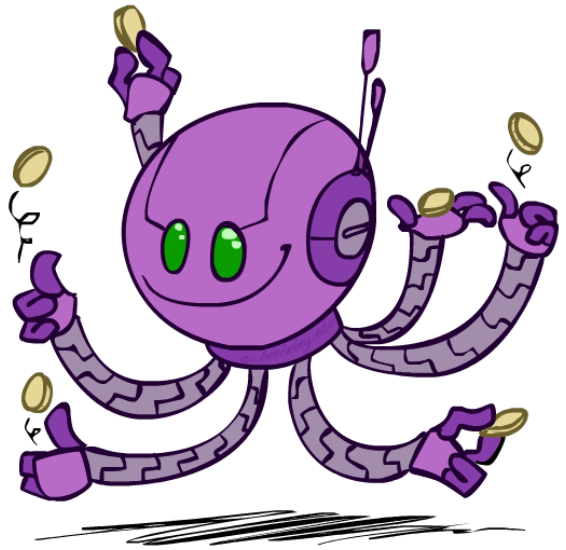
Example:



Graphical models

- Example: *N* independent coin flips



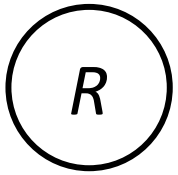


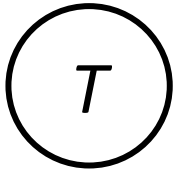
- No interaction between variables: absolute independence

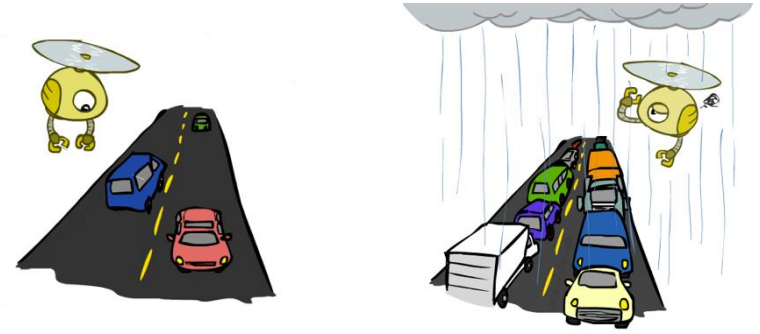
Graphical models

Example: Traffic

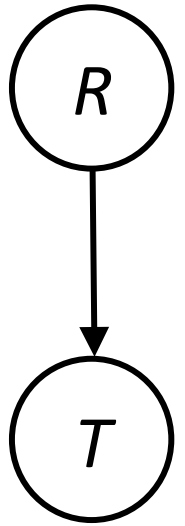
- Variables:
  - R: it rains
  - T: There is traffic
- Model 1: Independence





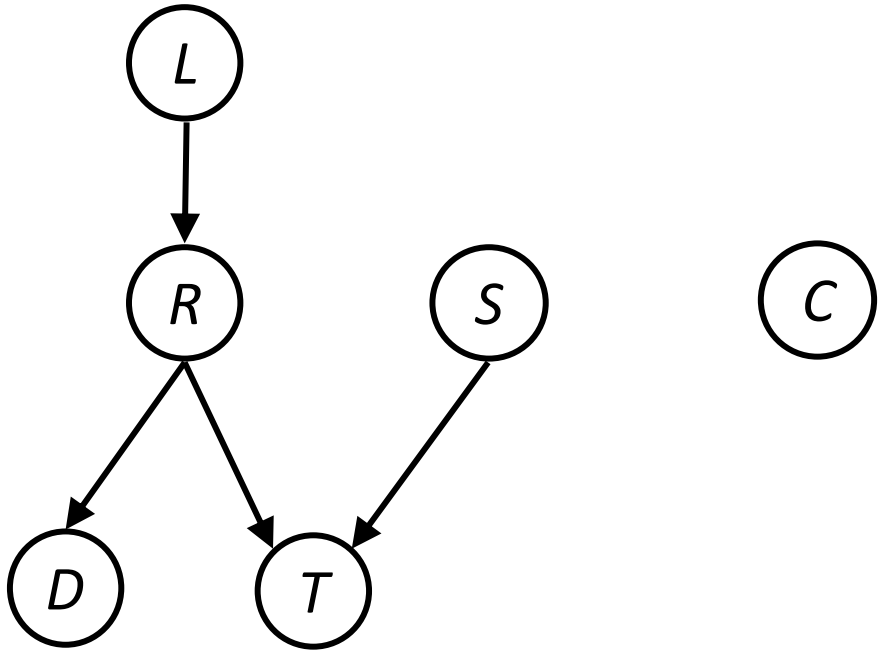


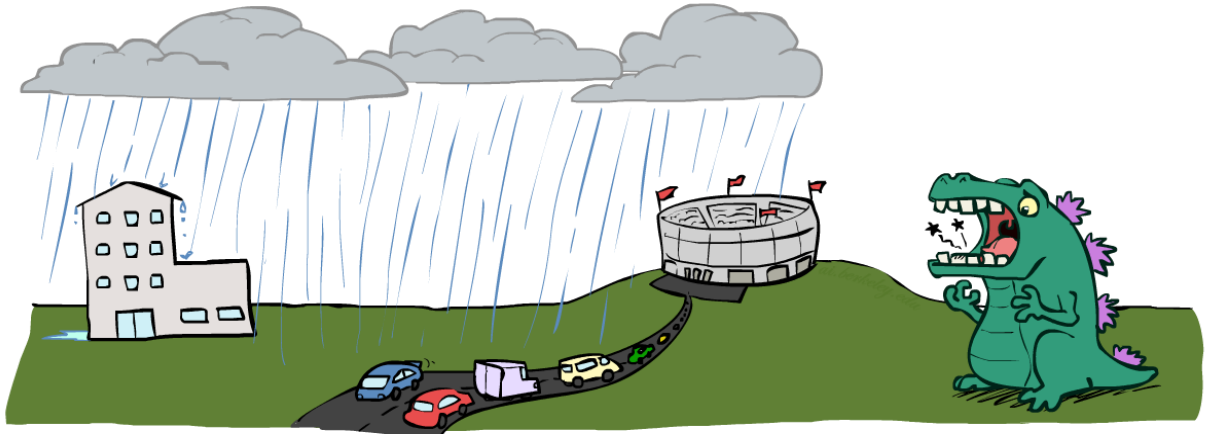
Model 2: Rain causes traffic



Graphical models

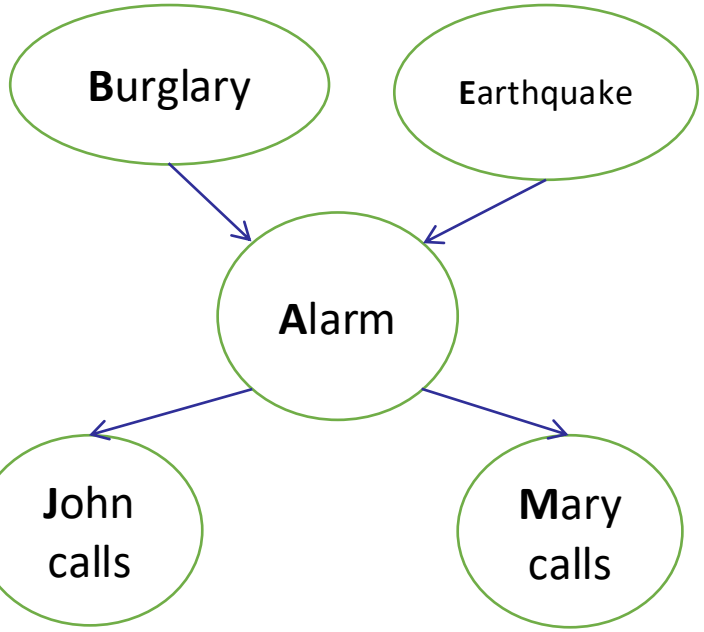
- Example: Traffic
- Variables:
  - T: Traffic
  - R: It rains
  - L: Low pressure
  - D: Roof drips
  - S: Sports match
  - C: Cavity

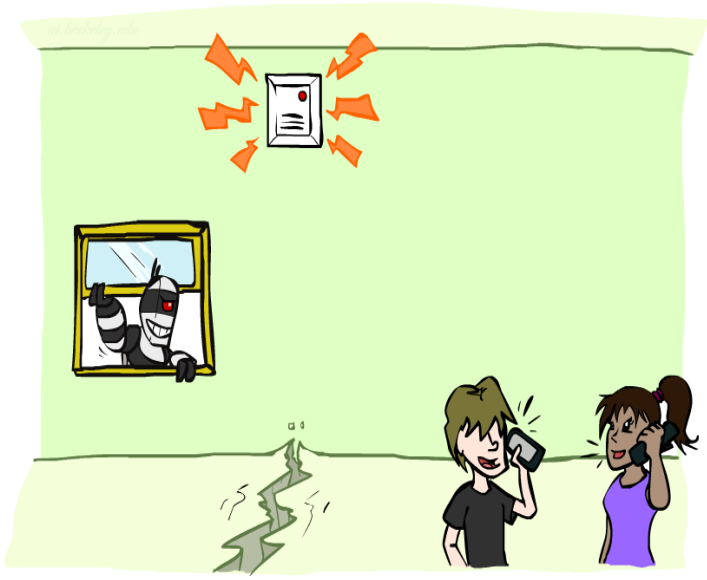




Graphical models

- Example: Alarm network
- Variables:
  - B: Burglary
  - A: Alarm goes off
  - M: Mary calls
  - J: John calls
  - E: Earthquake

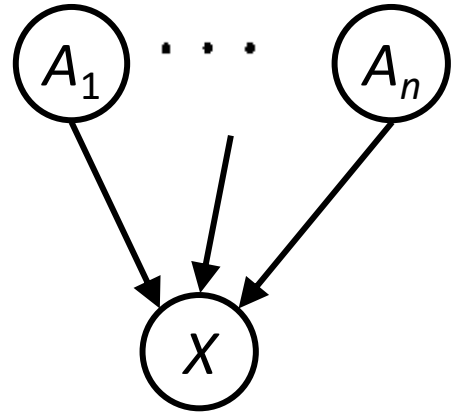




Probabilities in Bayesian Networks

A Bayesian network consists of:

- A *directed, acyclic* graph (no directed cycles), where each node represents a random variable X
- A local conditional probability distribution for each node:
  - $P(X|A_1, A_2, \ldots, A_n)$
  - i.e., a set of distributions $P(X|a_1, a_2, \ldots, a_n)$, one for each combination of possible values of X's *parent variables* A_1,...,A_n in the graph (i.e., the nodes for all incoming edges to X)
  - Stored in a conditional probability table (CPT)
  - Can be thought of as describing a noisy "causal" process

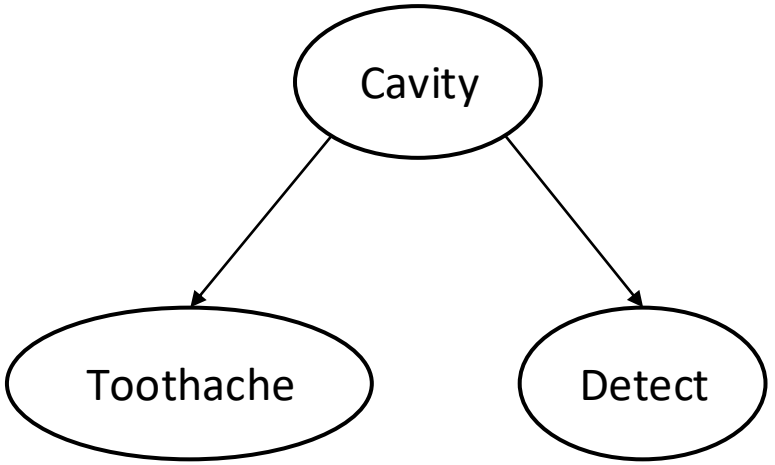


$P(X|A_1 \ldots A_n)$

Probabilities in Bayesian Networks

- Bayesian networks *implicitly* encode joint distributions as a product of local conditional distributions
- To compute the probability of a full assignment, multiply all the relevant conditional probabilities together:

$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i | parents(X_i))$$



- Example:

$$P(+cavity, +detect, -toothache)$$
$$= P(+cavity)\ P(-toothache|+cavity)\ P(+detect|+cavity)$$

Probabilities in Bayesian Networks

Why is this a valid definition of the joint distribution?

- Chain rule (valid for all distributions):

$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i | x_1 \ldots x_{i-1})$$

- Assume conditional independence in the Bayesian network (the nodes should be sorted topologically):

$$P(x_i | x_1, \ldots x_{i-1}) = P(x_i | \mathit{parents}(X_i))$$

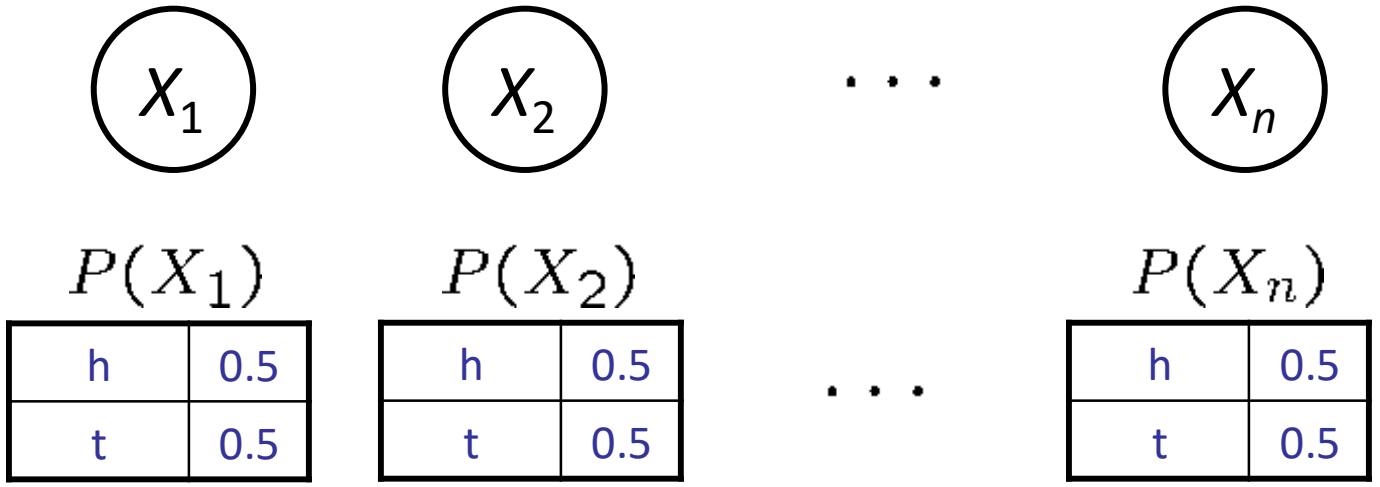
- Therefore:

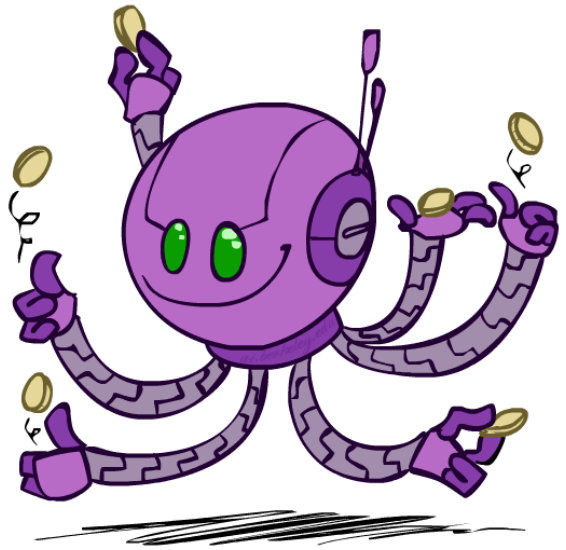
$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i | \mathit{parents}(X_i))$$

- Not every BN can represent every joint distribution: The topology enforces certain conditional independencies: It can only represent distributions consistent with those independencies

Probabilities in Bayesian Networks

- Example: *N* independent coin flips

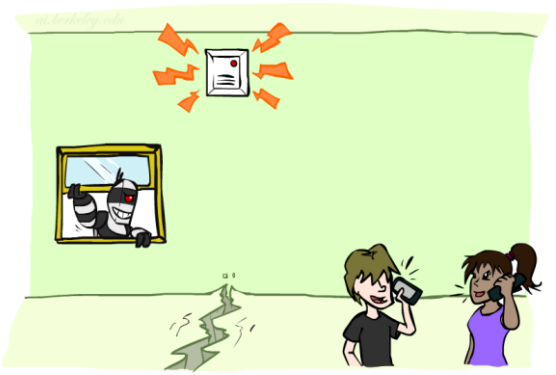




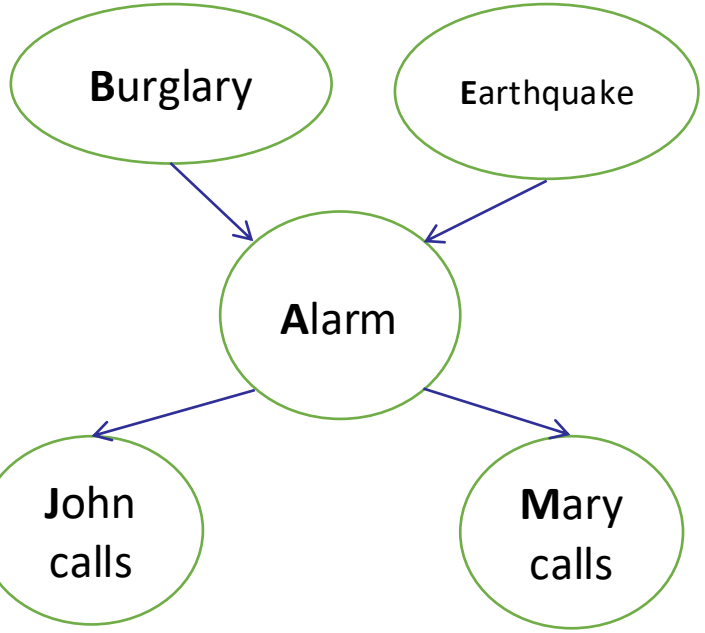
- Absolute independence: No arcs in the BN
- $P(h, h, t, h) = 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5$

Probabilities in Bayesian Networks

- Example: Alarm network $P(+b, -e, +a, -j, +m)$



B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

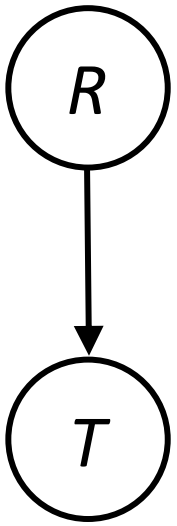
B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

Probabilities in Bayesian Networks

- Example: Traffic



$P(R)$

+r	1/4
-r	3/4

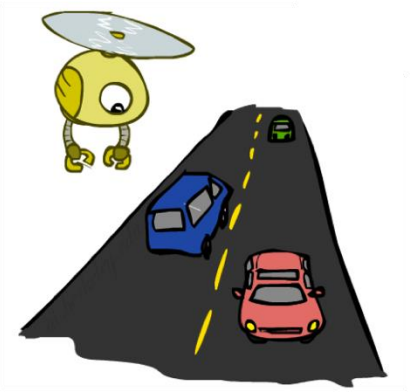
$P(T|R)$

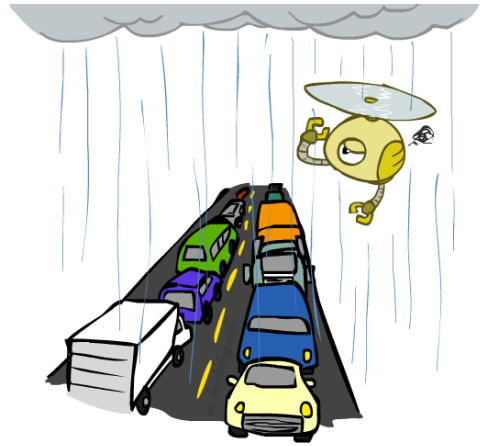
+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(+r, -t) = \frac{1}{4} \cdot \frac{1}{4}$$

$P(T, R)$

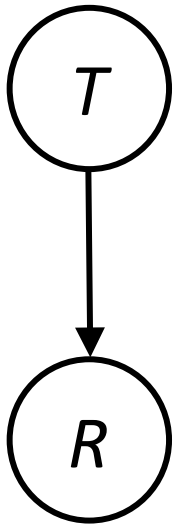
+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16





Probabilities in Bayesian Networks

- Example: Different topology, same joint distribution



$P(T)$

+t	9/16
-t	7/16

$P(R|T)$

+t	+r	1/3
	-r	2/3
-t	+r	1/7
	-r	6/7

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16



Probabilities in Bayesian Networks

BNs and causality:

- Arc direction does not necessarily imply causality
- When BNs do reflect the true causal patterns they are often simpler, easier to reason with, and easier to elicit from experts
- BNs do not need to be causal: In some domains there is not causality, then the arrows reflect correlation
- The topology really encodes conditional independence, which may happen to correspond to causal structure

Probabilities in Bayesian Networks

Size of a BN:

- Size of joint distribution over N Boolean variables: 2^N
- BN where nodes have up to k parents each: $O\left(N \cdot 2^{k+1}\right)$

Advantages over using full joint distribution:

- Substantial space saving
- Easier to elicit local conditional probability tables
- We'll see that it is also faster to do inference

Probabilities in Bayesian Networks

- A Bayesian network is an efficient encoding of a probabilistic model of a domain
- What we can do with a BN:
  - **Inference**: Given a fixed BN, what is $P(X \mid e)$?
  - **Representation**: Given a BN graph, what kinds of distributions can it encode?
  - **Modelling**: What BN is most appropriate for a given domain?
- When modelling a domain, we need to understand what assumptions are being made when designing the BN

